

CASE STUDY · GOVERNMENT VERTICAL · RENEWAL ANALYTICS

Predicting Government Churn, Winning Back Renewals

An end-to-end churn propensity system on GCP that gave account teams ranked outreach lists — and lifted renewal rates by 6–10%.

+6–10%

RENEWAL RATE LIFT

200+

GOV AGENCIES SERVED

3M+

USAGE RECORDS MODELED

Weekly

SCORING CADENCE

Renewals were rising in stakes — and outreach was running on instinct.

THE STAKES

We supported 200+ state and federal agencies using The Work Number for Medicaid and benefit verification. Several large contracts were approaching renewal under increasing competitive pressure.

HOW WE WERE OPERATING

Account teams prioritized renewal outreach using gut feel and basic usage reports. We had 3M+ historical usage records in BigQuery — but no systematic way to convert them into a ranked at-risk list.



Reactive renewals

Account teams found out a contract was at risk too late to do anything substantive about it.



High-value contracts at risk

Government renewals carry six- and seven-figure value — a single missed save outweighs the cost of any outreach.



Competitive pressure rising

Aggressive competitor moves into the gov vertical meant the cost of inaction was climbing each quarter.

Turn 3M usage records into a ranked renewal action list.

THE PROPOSAL

Build a churn propensity model that scores every active client weekly, ranks them by risk $\times$ contract value, and lands in the account team's CRM the next morning.



CHURN, DEFINED

Non-renewal at contract end — backed up by a usage/revenue-drop rule for ambiguous cases.

Co-validated with Sales Ops



PREDICTION WINDOW

Score 60 days before renewal using the prior 6 months of usage and account signals.

Point-in-time, leakage-free



PRIMARY METRIC

Renewal-rate lift in segments where account teams actioned the lists vs. baseline.

Business outcome, not AUC



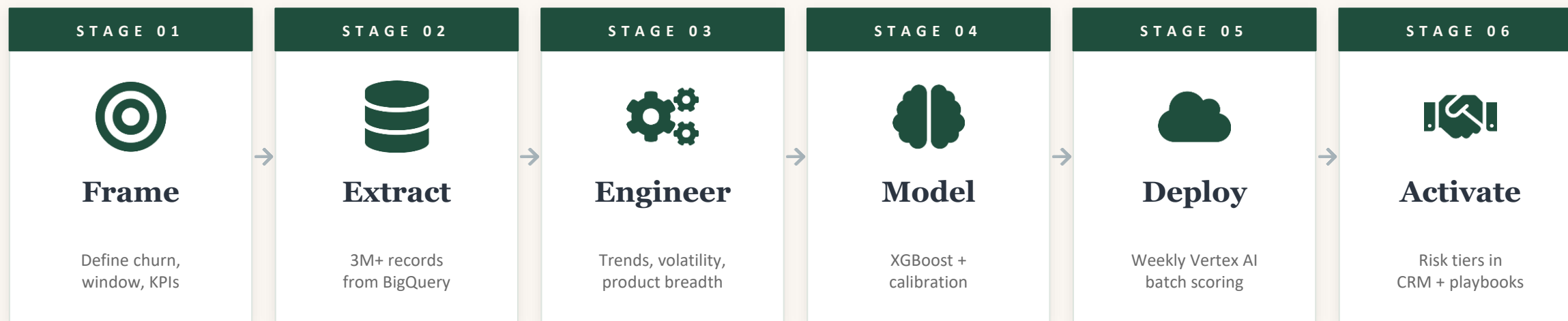
STAKEHOLDERS

Sales Ops for CRM integration; account managers for adoption; Pricing for treatment design.

Cross-functional from day one

Six stages — from BigQuery extract to weekly Vertex AI scoring.

I owned every stage end-to-end — and partnered with Sales Ops, account managers, and Pricing to make the scores actually move renewals.



WHY END-TO-END OWNERSHIP

A great model in a notebook moves nothing. Owning the full arc — from label definition to CRM playbook — was what made this land.

3M+ usage records, two tables, one carefully-defined label.

DATA SOURCES (BIGQUERY)



Usage events

3M+ rows: verifications, volumes, product mix, pricing per transaction



Contracts

Renewal dates, contract values, renewal flags, tenure



Account signals

Support tickets, escalations, payment issues, account-team interactions

CHURN — A TWO-PART LABEL

PRIMARY

Contract not renewed at end-of-term

Clean signal where renewal status was unambiguous — the bulk of cases.

BACKUP RULE

>40% revenue drop in next 60–90 days

For ambiguous renewal data — captures effective churn even without a contract event.



POINT-IN-TIME GUARDRAILS

Features use only signals observable up to the scoring date ($T = \text{renewal} - 60$ days). Label uses only post- T outcomes. Splits are time-based.

CLASS IMBALANCE · Churners are the minority — handled with XGBoost's `scale_pos_weight`, not naive resampling, so calibration stays honest.

Six feature families — every one a hypothesis about renewal risk.

Raw events → client-month aggregates → snapshot features at T-60 days. Built so account teams could read a feature and recognize the story.



Usage trends

3M and 6M usage averages, slope of change — captures decline before it shows up at renewal.



Volatility

Std-dev / mean of usage — turbulent accounts behave differently than steady decliners.



Product breadth

Count of products used — narrower footprints are easier for competitors to displace.



Price sensitivity

Usage response to past price changes — flags accounts that flinched on prior pricing moves.



Contract proximity

Days to renewal + tenure — the closer to renewal, the higher the leverage of acting now.



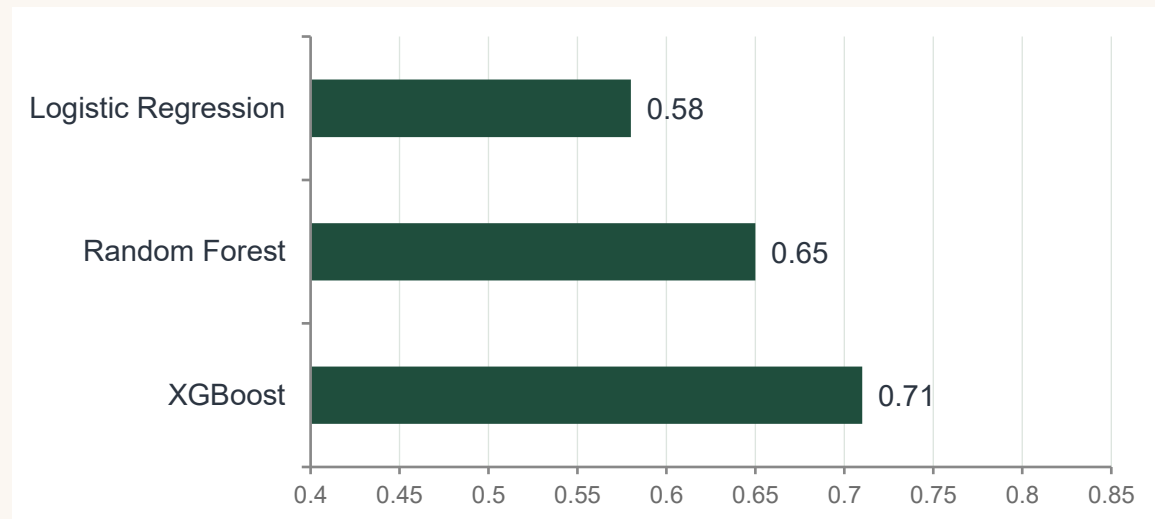
Support signals

Ticket volume, escalations, payment issues — operational friction that often precedes churn.

TRANSLATION · *Every feature mapped to a story an account manager could tell — "declining usage," "narrow product footprint," "escalating tickets."*

XGBoost classifier — chosen for non-linear interactions and imbalance handling.

BAKE-OFF — PR-AUC ON HOLDOUT



Price-sensitivity × usage-trend × support-friction interactions matter — and gradient boosting captured them better than the linear baseline.

WHY XGBOOST FOR THIS PROBLEM



Non-linear interactions, natively

Risk spikes when price changes meet usage drops meet support escalations — trees handle that without manual interaction terms.



Strong with imbalanced classes

`scale_pos_weight` tunes the loss for the minority churn class without distorting the data through resampling.



Handles mixed feature types

Numeric trends, categorical product flags, and ratio features all coexist — no heavy preprocessing required.



Drives feature attribution

SHAP values give per-account top drivers — exactly what account managers need at the call.

PRINCIPLE · XGBoost earned its complexity here — interactions and imbalance both made simpler models leave money on the table.

Calibrated probabilities, tiers that map to action — not just AUC.

HYPERPARAMETER TUNING

HYPERPARAMETER	RANGE	STRATEGY
n_estimators	200 → 600	early stopping
max_depth	3 → 7	grid
learning_rate	0.03 → 0.1	log-grid
scale_pos_weight	neg / pos	imbalance
min_child_weight	1 → 10	stability
calibration	isotonic	post-fit

Time-based CV — older months train, newer months validate. Held-out test set for the final unbiased read.

CALIBRATED TIERS

Isotonic calibration so $p = 0.7$ actually means ~70% — then tiered for the business.

HIGH $p \geq 0.70$	Exec sponsor + pricing review
MEDIUM $0.40 \leq p < 0.70$	Proactive check-in + training
LOW $p < 0.40$	Standard renewal motion

PRIORITIZATION · $\text{expected_risk_value} = \text{churn_prob} \times \text{contract_value}$ — saves the next dollar where it matters most.

BigQuery + Vertex AI — a self-updating weekly pipeline.



Self-updating by design

Scheduled BQ queries refresh features. Vertex AI retrains weekly. Scores land back in BigQuery automatically — no manual handoffs in the steady state.



Production guardrails

Monitoring on data freshness, score-distribution drift, feature drift, and pipeline run-success — so silent regressions get caught early.

STACK · BigQuery (data + features + scores) · Vertex AI (training + batch scoring) · Looker / CRM (consumption).

A score is just a number until account teams act on it.

The model wasn't going to move renewals by itself. The lift came from making it obvious what to do with each tier — and putting the answer in the tools account teams already lived in.



Embedded in CRM

Partnered with Sales Ops to surface risk tiers and top drivers directly inside the CRM views and renewal dashboards account managers already used.



Enablement sessions

Walked account teams through what the score meant, what drove it, and how to read top features. The score was useless until people understood it.



Tier-specific playbooks

Co-designed with Pricing: high-risk + high-value got exec sponsorship and pricing review; medium-risk got proactive check-ins; low-risk stayed in the standard motion.

Renewal rates moved where the lists were actioned.

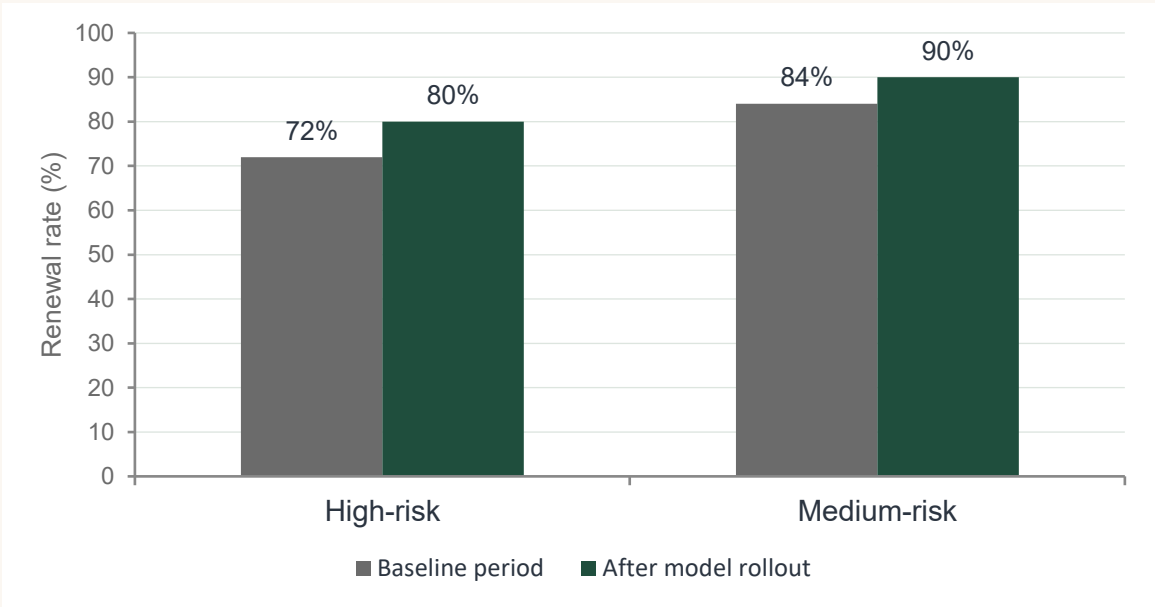
HOW WE MEASURED

Compared renewal rates in segments where account teams actively used the lists vs. matched baseline segments and prior periods.

WHAT MOVED

- Renewal rate — primary outcome
- Save rate within high-risk tier specifically
- Outreach efficiency — contacts per save
- Pipeline rhythm — weekly operating cadence

RENEWAL RATE — TARGETED vs. BASELINE



Illustrative — directional values shown for confidentiality.



HEADLINE IMPACT

+6–10% renewal rate improvement in segments where account teams actioned the lists.

The scoreboard.



RENEWAL RATE LIFT

+6–10%

in segments where account teams actioned the lists



GOV AGENCIES SERVED

200+

state and federal customers across the portfolio



USAGE RECORDS MODELED

3M+

BigQuery-resident, ~6 months of context per scoring run



AUTOMATED CADENCE

Weekly

Vertex AI retraining + BigQuery writes — zero manual touches steady-state

QUALITATIVE WINS

Weekly renewal pipeline became a standard operating rhythm · Account teams shifted from reactive to proactive outreach · Pipeline reusable across other Equifax verticals.

What worked, what I'd extend next.

WHAT WORKED



Building the prioritization, not just the score

Sorting by churn-prob $\times$ contract-value gave account teams a list they could act on Monday morning — not just a probability column.



Calibrating probabilities, not just AUC

Tiers were only useful because $p = 0.7$ actually meant $\sim 70\%$. Calibration earned the right to be tiered.



Co-built playbooks with Pricing & Sales Ops

Tier $\times$ treatment $\times$ outcome was designed together — that's why the lists got worked, not ignored.

WHAT I'D EXTEND NEXT



Move from churn prediction to uplift

Prediction tells you who's risky. Uplift tells you who's persuadable — directly optimizes the outreach budget.



Feed campaign outcomes back as features

Past treatment-and-response data closes the loop and makes each model generation smarter than the last.



Drift-triggered retraining

Move beyond weekly cadence to retraining triggered by feature or score drift — faster reaction to portfolio shifts.

From gut-feel renewals to a weekly operating rhythm.

01 Prioritize, don't just predict.

A churn probability is interesting. A list ranked by expected revenue at risk is a Monday-morning action item — and that's what changes outcomes.

02 Calibrate before you tier.

Tiers only mean what the probabilities mean. Isotonic calibration is what turned scores into something Sales could trust.

03 End-to-end ownership beats handoffs.

Owning data, model, deployment, and adoption together was the difference between an interesting analysis and a system the business now runs on.

Thank you · Questions?